

Unit-2

Propositional Logic

Propositional or st sentence $\rightarrow$ An Expression
Consisting of some symbols, letters and words
is called a Sentence if it is true or false.

Ex* Following are statement (or propositional)

Declarative Sentence

Truth Table

- | | | |
|----|---|-------|
| 1) | $\{x: x^2 = 36\} = \{6, -6\}$ | True |
| 2) | $3 > 9$ | False |
| 3) | The capital of Madhya Pradesh is Bhopal | True |
| 4) | $6 + 4 = 10$ | True |
| 5) | $9 < 6$ | False |
| 6) | Jaipur is capital of Rajasthan | True |
| 7) | Blood is red | True |
| 8) | Mumbai is in America | False |

$\rightarrow$ "Wish you happy life" is not a Proposition
because Truth or False is not certain.

Statement letters or Sentence Variable $\rightarrow$

Definition $\rightarrow$ The Symbols, which are used to represent statements are called Statement letters or Sentence Variables.

To represent statement usually the letters $P, Q, R, \dots, p, q, r, \dots$ etc, are used.

For ex. if the statement "Bhopal is the Capital of Madhya Pradesh" is denoted by the letter 'p' then in Mathematics it is written as follows:

$p =$ Bhopal is the capital of Madhya Pradesh.

Logical Connectives or Sentence Connectives $\rightarrow$

Logical Connectives or Sentence Connectives are the words or symbols used to combine two sentences or statements to form a compound sentence or compound statement.

Connective Word	Name of Connective	Symbol	Rank.
Not	Negation	$\sim$	1
and	Conjunction	$\wedge$	2
or	Disjunction	$\vee$	3
if ... then	Conditional	$\rightarrow$ or $\Rightarrow$	4
iff or if and only if	Bi-Conditional	$\leftrightarrow$ or $\Leftrightarrow$	5

Types of Proposition / Sentences →

1) Simple Proposition → The Proposition having one Subject and one Predicate is called a Simple Proposition.

Ex → ① This flower is pink.

② Every even no. is divisible by 2.

2) Compound Proposition → Two or more Simple Propositions when combined by various connectivities into a single composite sentence is called Compound Proposition.

Ex → ① The earth is round and revolves around the sun.

② A triangle is equilateral iff its three sides are equal.

Logical Connectives →

The Particular words and Symbols used to join two or more Proposition into a single Composite form or Compound Proposition are called logic Connectives.

Logical Connectives Words	Symbol	uses.
And / Conjunction / Join	$\wedge$	$P \wedge Q$
or / Disjunction / meet	$\vee$	$P \vee Q$
Negation	$-$ or $\sim$	$\sim P$
Equivalent	$\longleftrightarrow$	$P \longleftrightarrow Q$
Conditional "if ... then ..."	$\Rightarrow$	$P \Rightarrow Q$
Biconditional "if and only if"	$\Longleftrightarrow$	$P \Longleftrightarrow Q$
NAND (NOT + AND)	$\uparrow$	$P \uparrow Q$
NOR (NOT + OR)	$\downarrow$	$P \downarrow Q$
XOR	$\oplus$	$P \oplus Q$

Truth Value

If any proposition is true then its truth value is denoted by T and if the proposition is false then its truth table value is denoted by F

<u>Ex</u> $\Rightarrow$ (1)	1 is less than 3.	T
(2)	14 is odd no.	F
(3)	9 is prime no.	F

Ex $\Rightarrow$ (1) Which of the following statements are true and which are false.

1) $16 + 9 = 4^2 + 3^2$
True, Since $25 = 25$

2) $9 \times 7 = 21 \times 3$
True, Since $63 = 63$

3) $\{2, 3\} \subset \{2, 4, 6\}$
False, Since $3 \notin \{2, 4, 6\}$

4) $5 \in \{1, 3, 5\}$
True, Since 5 is an element of the set $\{1, 3, 5\}$

5) $5 + x = 9$, $x = ?$
 $x = 4$,

Ex $\rightarrow$ (2) If $p =$ he is poor, $q =$ he is laborious, then write down the following statement in symbols. (i.e. in the language of logic)

- i) He is poor and laborious.
 $\rightarrow p \wedge q$
- ii) He is poor but is not laborious.
 $\rightarrow p \wedge \sim q$
- iii) It is false that he is poor or laborious.
 $\rightarrow \sim(p \vee q)$
- iv) It is false that he is not poor or is laborious.
 $\rightarrow \sim(\sim p \vee q)$
- v) Neither he is poor nor he is laborious.
 $\rightarrow \sim p \wedge \sim q$
- vi) He is poor or is not poor and is laborious.
 $\rightarrow p \vee (\sim p \wedge q)$
- vii) It is not true that he is p. not poor or is not laborious.
 $\rightarrow \sim(\sim p \vee \sim q)$

Ex $\rightarrow$ (3) Let $p =$ it is cold, $q =$ it is raining. Give a simple verbal sentence which describes each of the following statements.

- i) $\sim p$
 $\rightarrow$ It is not cold
- ii) $p \wedge q$
 $\rightarrow$ It is cold and raining

iii) $p \vee q$

So $\rightarrow$ It is cold or it is raining.

iv) $q \Leftrightarrow p$

It is raining if and only if it is cold.

v) $p \Rightarrow \neg q$

If it is cold, then it is not raining.

vi) $q \vee \neg p$

It is raining or it is not cold.

vii) $\neg p \wedge \neg q$

It is not cold and it is not raining.

viii) $p \Leftrightarrow \neg q$

It is cold if and only if it is not raining.

ix) $\neg \neg q$

It is not true that it is not raining.

x) $(p \wedge q) \Rightarrow p$

If it is cold and not raining, then it is cold.

xi) $\neg \neg p$

It is not true that it is not cold.

xii) $(p \wedge \neg q) \Rightarrow q$

If it is cold and not raining, then it is raining.

Basic Logical Operations →

1) Conjunction ($\wedge$) → Suppose p and q are two statements

When two statements p, q are combined by using the word "and", then we get a new statement represented by $p \wedge q$. The resulting statement $p \wedge q$ is read as " p and q " or " p meet q ". The statement $p \wedge q$ so obtained is called the Conjunction of p and q .

Truth Table

P	q	$P \wedge q$
T	T	T
T	F	0
F	T	0
F	F	0

Ex → ① Delhi is in india and $2+2=4$ T

② Delhi is in india and $2+2=5$ F

③ Delhi is in ~~in~~ Russia and $2+2=4$ F

2) Disjunction ($\vee$) → Suppose p and q are two statements

When two statements p, q are joined by using the word "or" then we get a new statement represented by $p \vee q$. The resulting statement $p \vee q$ is read as " p or q ", " p join q ".

The statement $p \vee q$ so obtained is called the disjunction of p and q .

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

- Ex →
- ① Delhi is in India or $2+2=4$ T
 - ② Delhi is in India or $2+2=5$ T
 - ③ Delhi is in Russia or $2+2=4$ T
 - ④ Delhi is in Russia or $2+2=6$ F

3) Negation ($\sim$) → Let p be a statement. The negation of the statement p is obtained by putting the word "not" and it is represented by " $\sim p$ " or by ' $\bar{p}$ ', $\sim p$ is read as "not p ".

p	$\sim p$
T	F
F	T

Ex ① p = The flower is pink
 $\sim p$ = The flower is not pink.

4) Conditional ($\Rightarrow$ or $\rightarrow$) $\rightarrow$ Let p and q any two statements. The statement of the type "if p then q " is called Conditional statement and is represented by $(p \Rightarrow q)$ or $(p \rightarrow q)$. It is read as 'p implies q'.

p	q	$p \rightarrow q$
T	T	T
F	F	T
T	F	F
F	T	T

5) Bi-conditional ($\Leftrightarrow$ or $\leftrightarrow$) Let p and q be two statements. The statement of the type ' $p \Rightarrow q$ and $q \Rightarrow p$ ' i.e. ' $p \Rightarrow q \wedge q \Rightarrow p$ ' is called bi-conditional statement and it is represented by ' $p \Leftrightarrow q$ ' or by ' $p \leftrightarrow q$ '.

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

6) NOR ($\downarrow$) $\rightarrow$ Let P and Q be 2 statement.
 The statement of the type $P \downarrow Q$, read as "neither P nor Q" is called joint denial or NOR.
 In other words NOR is the "negation of OR of 2 statement" True. $(P \downarrow Q) = \sim(P \vee Q)$

(It means that $P \downarrow Q$ is true only when P and Q both are false)

Truth Table $\rightarrow$

P	Q	$P \downarrow Q$
T	T	F
T	F	F
F	T	F
F	F	T

7) NAND ($\uparrow$) $\rightarrow$ The connective NAND is denoted by the symbol $\uparrow$ and is defined as "negation of AND of two statement and it is denoted $P \uparrow Q$ ".
 Thus $P \uparrow Q = \sim(P \wedge Q)$
 ($P \uparrow Q$ is false only is the case when both P and Q are true.)

Truth Table

P	Q	$P \uparrow Q$
T	T	F
T	F	T
F	T	T
F	F	T

8) XOR $\oplus$ $\rightarrow$ The XOR of 2 statement is True if and only if one of the two statement is true and it is denoted by $P \oplus Q$

Truth Table

P	Q	$P \oplus Q$
T	T	F
T	F	T
F	T	T
F	F	F

Tautology $\rightarrow$

A tautology is a Proposition which is true for all truth values of its Sub Propositions or Components.

Contradiction $\rightarrow$

A Contradiction is a Proposition which is always false for all truth values of its Propositions or Components.

Ex → ③

Prove that $p \wedge q \Rightarrow q$ is tautology

Sol →

Truth table for $p \wedge q \Rightarrow q$

p	q	$p \wedge q$	$p \wedge q \Rightarrow q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

Ex →

Prove that $p \wedge q \Rightarrow p \vee q$ is tautology

Ex → ④

Prove that $p \wedge q \Rightarrow p \vee q$ is tautology

Sol →

Truth table for $p \wedge q \Rightarrow p \vee q$

p	q	$p \wedge q$	$p \vee q$	$p \wedge q \Rightarrow p \vee q$
T	T	T	T	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	T

Ex ⑤ Prove that $\sim\{p \wedge (\sim p)\}$ is a tautology

Sol →

p	$\sim p$	$p \wedge (\sim p)$	$\sim\{p \wedge (\sim p)\}$
T	F	F	T
F	T	F	T

Ex → ⑥ Prove that $(p \Leftrightarrow q) \Leftrightarrow (p \Rightarrow q) \wedge (q \Rightarrow p)$ is tautology.

Sol →

p	q	$p \Leftrightarrow q$	$p \Rightarrow q$	$q \Rightarrow p$	$(p \Rightarrow q) \wedge (q \Rightarrow p)$	$(p \Leftrightarrow q) \Leftrightarrow (p \Rightarrow q) \wedge (q \Rightarrow p)$
T	T	T	T	T	T	T
T	F	F	F	T	F	T
F	T	F	T	F	F	T T
F	F	T	T	T	T	T

Ex → ⑦ Prove that $(p \Rightarrow q) \Leftrightarrow (\sim p \vee q)$ is a tautology

Sol →

p	q	$\sim p$	$p \Rightarrow q$	$\sim p \vee q$	$p \Rightarrow q \Leftrightarrow (\sim p \vee q)$
T	T	F	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

Ex → ⑧

Ex - (9) Prove that $p \wedge (\sim p)$ is a Contradiction.

Sol.

p	$\sim p$	$p \wedge (\sim p)$
T	F	F
F	T	F

Ex - (10) Prove that $p \Leftrightarrow (\sim p)$ is a Contradiction.

p	$\sim p$	$p \Leftrightarrow (\sim p)$
T	F	F
F	T	F

Ex - (11) Prove that $(p \vee q) \wedge (\sim p) \wedge (\sim q)$ is a Contradiction

Sol.

p	q	$p \vee q$	$\sim p$	$\sim q$	$(p \vee q) \wedge (\sim p)$	$(p \vee q) \wedge (\sim p) \wedge (\sim q)$
T	T	T	F	F	F	F
T	F	T	F	T	F	F
F	T	T	T	F	T	F
F	F	F	T	T	F	F

Ex - (12) Prove that $(\sim p \wedge q) \vee (p \wedge \sim q)$ is a Contradiction.

p	q	$\sim p$	$\sim q$	$\sim p \wedge q$	$p \wedge \sim q$	$(\sim p \wedge q) \vee (p \wedge \sim q)$
T	T	F	F	F	F	F
T	F	F	T	F	T	T
F	T	T	F	T	F	T
F	F	T	T	F	F	F

Ex → Prove that the Proposition $p \vee \sim(q \wedge r)$ and $(p \vee \sim q) \vee \sim r$ are equivalent.

Sol →

P	q	r	$q \wedge r$	$\sim(q \wedge r)$	$p \vee \sim(q \wedge r)$
T	T	T	T	F	T
F	T	T	T	F	F
T	F	F	F	T	T
F	F	F	F	T	T
T	T	F	F	T	T
T	F	T	F	T	T
F	T	F	F	T	T
F	F	T	F	T	T

Logic Equivalence →

Two statements or (Propositions) are called logically equivalent if the truth values of both the statements or (Propositions) are ~~not~~ always identical.

If two statements P and Q are logically equivalent, then these are represented by ' $P \equiv Q$ '. If $P \equiv Q$ then $P \Rightarrow Q$ is a tautology.

Show that $P \Rightarrow (Q \Rightarrow R) \equiv (P \wedge Q) \Rightarrow R$

P	Q	R	$Q \Rightarrow R$	$P \Rightarrow (Q \Rightarrow R)$	$(P \wedge Q)$	$(P \wedge Q) \Rightarrow R$
T	T	T	T	T	T	T
T	T	F	F	F	T	F
T	F	T	T	T	F	T
T	F	F	T	T	F	T
F	T	T	T	T	F	T
F	T	F	F	T	F	T
F	F	T	T	T	F	T
F	F	F	T	T	F	T

Ex: Show that $\sim(P \Rightarrow Q) \equiv \{P \wedge (\sim Q)\}$

Ans:

P	Q	$\sim Q$	$P \Rightarrow Q$	$\sim(P \Rightarrow Q)$	$P \wedge (\sim Q)$
T	T	F	T	F	F
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	F	F

~~$P \Rightarrow Q$~~

Converse, Inverse and Contrapositive $\rightarrow$

We can form some new Conditional Statements starting with a Conditional Statement $P \rightarrow Q$

there are three related conditional statements that occur so often that they have special names.

- 1) Converse $[P \rightarrow Q \text{ Converse } Q \rightarrow P]$
- 2) Inverse $[P \rightarrow Q \text{ Inverse } \sim P \rightarrow \sim Q]$
- 3) Contrapositive $[P \rightarrow Q \text{ Contrapositive } \sim Q \rightarrow \sim P]$

Ex $\rightarrow$ What are the Contrapositive, the Converse and the Inverse of the Conditional statement. "The home team wins whenever it is raining?"

$P =$ It is raining

$Q \Rightarrow$ Home team wins.

$P \rightarrow Q$, If it is raining, then ^{the} home team wins.

Converse $\rightarrow [Q \rightarrow P]$ "If the home team wins, then it is raining."

Inverse $\rightarrow [\sim P \rightarrow \sim Q]$ "If it is not raining, then the home team does not win."

Contrapositive $\rightarrow [\sim Q \rightarrow \sim P]$ "If the home team does not win, then it is not raining."

Example 2. (a) Obtain the converse, inverse and contrapositive of the conditional statement $p \rightarrow q$. [R.G.P.V. Dec. 2002]

Solution. If p and q are two propositions then the converse, inverse and contrapositive of $p \rightarrow q$ are as follows :

(i) Converse implication is $q \rightarrow p$.

(ii) Inverse implication is $\sim p \rightarrow \sim q$.

(ii) Contrapositive implication is $\sim q \rightarrow \sim p$.

Example 2. (b) Construct converse, inverse and contrapositive of the direct statement 'if $4x - 2 = 10$ then $x = 3$ '.

Solution. Let $p \equiv 4x - 2 = 10$, $q \equiv x = 3$. Thus the statement 'if $4x - 2 = 10$ then $x = 3$ ' is the direct implication $p \Rightarrow q$.

(i) **Converse.** Converse of ' $p \Rightarrow q$ ' is ' $q \Rightarrow p$ ', hence the converse of given statement is as follows :

"If $x = 3$ then $4x - 2 = 10$ ".

(ii) **Inverse.** The inverse of ' $p \Rightarrow q$ ' is ' $(\neg p) \Rightarrow (\neg q)$ ', hence the inverse of the given statement is as follows :

"If $4x - 2 \neq 10$ then $x \neq 3$ ".

(iii) **Contrapositive.** The contrapositive of ' $p \Rightarrow q$ ' is ' $(\neg q) \Rightarrow (\neg p)$ ', hence the contrapositive of the given statement is as follows :

"If $x \neq 3$ then $4x - 2 \neq 10$ ".

Example 3. Write the converse, inverse and contrapositive of the following direct statements :

(a) If $ABCD$ is a square then $ABCD$ is a rectangle.

(b) When sun rises then it is morning.

Solution (a) Let $p \equiv ABCD$ is a square,

$q \equiv ABCD$ is a rectangle.

$\therefore$ The given statement is the direct implication ' $p \Rightarrow q$ '.

(i) **Converse.** Since the converse of ' $p \Rightarrow q$ ' is ' $q \Rightarrow p$ ' so the converse of given statement is :

'If $ABCD$ is a rectangle then $ABCD$ is a square'.

(ii) **Inverse.** Since inverse of ' $p \Rightarrow q$ ' is ' $(\neg p) \Rightarrow (\neg q)$ ' so the inverse of given statement is :

'If $ABCD$ is not a square then $ABCD$ is not a rectangle'.

(iii) **Contrapositive.** Since contrapositive of ' $p \Rightarrow q$ ' is ' $(\neg q) \Rightarrow (\neg p)$ ', so the contrapositive of the given statement is :

'If $ABCD$ is not a rectangle then $ABCD$ is not a square'.

(b) Assuming p and q as above in (a), the given statement has :

(i) **Converse.** 'When it is morning then sun rises'.

(ii) **Inverse.** 'When sun does not rise then it is not morning'.

(iii) **Contrapositive.** 'When it is not morning then sun does not rise'.

Example 4. Let $p \equiv$ it is rainy season, $q \equiv$ the mango is delicious, construct the following statements :

(a) $p \Rightarrow q$,

(b) its converse,

(c) its inverse,

(d) its contrapositive,

(e) its negation.

Solution. (a) If it is rainy season, then the mango is delicious.

(b) If the mango is delicious, then it is rainy season.

(c) If it is not rainy season, then the mango is not delicious.

(d) If the mango is not delicious, then it is not rainy season.

(e) It is rainy season and mango is not delicious.

Example 5. If the truth value of $p \Leftrightarrow q$ is (i) F , (ii) T , then find the truth value of $p \vee q$.

Solution. We know that the truth value of $p \Leftrightarrow q$ is F , when either one of p and q has truth value F and other T .

Truth table

$p \Leftrightarrow q$	p	q	$p \vee q$
F	T	F	T
	F	T	T

Since the entries in the column under $p \vee q$ are of only T 's, hence the truth value of $p \vee q$ is 'True' (T).

(ii) The truth value of $p \Leftrightarrow q$ is T when its both components p and q has the same truth values (i.e., T or F).

Truth table

$p \Leftrightarrow q$	p	q	$p \vee q$
T	T	T	T
	F	F	F

From truth table we see that the truth value of $p \vee q$ is sometimes T and sometimes F , hence the truth value of $p \vee q$ cannot be decided.

Example 6. If the truth value of $p \vee q$ is F , then decide the truth value of $p \Leftrightarrow q$.

Solution. We know that the truth value of $p \vee q$ is F , only in the condition 'when p and q both have truth values F '. Hence the truth value of $p \Leftrightarrow q$ is T (see following truth table).

$p \vee q$	p	q	$p \Leftrightarrow q$
F	F	F	T

Example 7. If $P \Leftrightarrow Q$ is true, then determine the truth value of $P \vee (\sim Q)$.

Solution. Truth table

$P \Leftrightarrow Q$	P	Q	$\sim Q$	$P \vee (\sim Q)$
T	T	T	F	T
	F	F	T	T

Hence the truth value of $P \vee (\sim Q)$ is 'True' (T).

Example 8. If P is a statement such that for any statement Q , the statement $P \wedge Q$ is false, then what do you conclude about the truth value of P ?

Solution. We know that the truth value of $P \wedge Q$ is 'True' when truth values of P and Q both are 'True', and in all other cases the truth value of $P \wedge Q$ is 'False'. Therefore, in the given condition, the truth value of P is given by the third column of the following truth table :

$P \wedge Q$	Q	P
F	F	T
F	T	F
F	F	F

Example 9. If P is a statement such that for any statement Q the statement $P \vee Q$ is true, then what do you conclude about the truth value of P ?

Solution. The truth value of P is given by the third column of the following truth table :

$P \vee Q$	Q	P
T	T	F
T	F	T
T	T	T

Solution. Truth table for $(P \Rightarrow Q \wedge R) \vee (\sim P \wedge Q)$:

P	$Q \wedge R$	$P \Rightarrow (Q \wedge R)$	$\sim P$	$\sim P \wedge Q$
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Solution.

P	Q	R	$Q \wedge R$	$P \Rightarrow (Q \wedge R)$	$\sim P$	$\sim P \wedge Q$	$(P \Rightarrow Q \wedge R) \vee (\sim P \wedge Q)$
T	T	T	T	T	F	F	T
T	T	F	F	F	F	F	F
T	F	T	F	F	F	F	F
T	F	F	F	F	F	F	F
T	T	T	T	T	T	T	T
F	T	F	F	T	T	T	T
F	T	T	T	T	T	F	T
F	F	T	F	T	T	F	T
F	F	F	F	T	T	F	T

$$(\sim B) \wedge (A \Rightarrow B) \Rightarrow (\sim A).$$

Solution. Truth table for given statement :

A	B	$\sim A$	$\sim B$	$A \Rightarrow B$	$\sim B \wedge (A \Rightarrow B)$	$(\sim B) \wedge (A \Rightarrow B) \Rightarrow (\sim A)$
T	T	F	F	T	F	T
T	F	F	T	F	F	T
F	T	T	F	T	F	T
F	F	T	T	T	T	T

Hence $(\sim B) \wedge (A \Rightarrow B) \Rightarrow (\sim A)$ is a tautology.

Example 12. If $p \equiv \text{Ram is beautiful,}$
 $q \equiv \text{Ram is mixable,}$
 $r \equiv \text{His friends like Ram,}$

then write the following statements in language :

- (a) $(p \Rightarrow q) \vee (p \Rightarrow r)$,
(b) $p \Rightarrow (q \vee r)$.

Examine, are the above statement equivalent?

Solution. (a) If Ram is beautiful then either Ram is mixable or his friends Ram.

(b) If Ram is beautiful then he is mixable or his friends like him.

Combined truth table for (a) and (b) :

p	q	r	$p \Rightarrow q$	$p \Rightarrow r$	$(p \Rightarrow q) \vee (p \Rightarrow r)$	$q \vee r$	$p \Rightarrow (q \vee r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	F	T

From above truth table, we see that the entries in sixth and eighth column are identical. Hence the statements given by (a) and (b) above are logically equivalent.

Example 13. Prepare the tables of the following statements :

(i) $(p \Leftrightarrow q) \wedge (r \vee q)$.

(ii) $(p \vee q) \wedge (\sim r) \Rightarrow q$.

Solution. (i) Truth table of $p \Leftrightarrow q \wedge (r \vee q)$:

p	q	r	$p \Leftrightarrow q$	$r \vee q$	$(p \Leftrightarrow q) \wedge (r \vee q)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	F	F	F
F	T	T	F	T	F
F	T	F	F	T	F
F	F	T	T	T	T
F	F	F	T	F	F

(ii) Truth table of $(p \vee q) \wedge (\sim r) \Rightarrow q$:

p	q	$\sim r$	$p \vee q$	$(p \vee q) \wedge (\sim r)$	$(p \vee q) \wedge (\sim r) \Rightarrow q$
T	T	T	T	T	T
T	T	F	T	F	T
T	F	T	T	T	F
T	F	F	T	F	T
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	F	F	T
F	F	F	F	F	T

Example 14. Prove by truth table that the following formulae are tautologies :

(i) $(\sim q \Rightarrow \sim p) \wedge (q \Rightarrow p) \Rightarrow (p \Leftrightarrow q)$, (ii) $(p \Rightarrow q) \vee (r \Rightarrow p)$.

(iii) $(p \Leftrightarrow q \wedge r) \Rightarrow (\sim r \Rightarrow \sim p)$, (iv) $(p \Rightarrow q \wedge r) \Rightarrow (\sim r \Rightarrow \sim q)$.

Solution. (i) Let $\sim q \Rightarrow \sim p$, $q \Rightarrow p$, $p \Leftrightarrow q$ be respectively denoted by A , B , C . The truth table of given statement is as follows :

p	q	$\sim p$	$\sim q$	A	B	$A \wedge B$	C	$(A \wedge B) \Rightarrow C$
T	T	F	F	T	T	T	T	T
T	F	F	T	F	T	F	F	T
F	T	T	F	T	F	F	F	T
F	F	T	T	T	T	T	T	T

Since all entries of last column are T . Hence the given formula is a tautology.

Proposition

(ii)

p	q	r	$p \Rightarrow q$	$r \Rightarrow p$	$(p \Rightarrow q) \vee (r \Rightarrow p)$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	F	T	T
T	F	F	F	T	T
F	T	T	T	F	T
F	T	F	T	T	T
F	F	T	T	F	T
F	F	F	T	T	T

Since all entries of the last column are T . Hence the given formula is a tautology.

(iii)

p	q	r	$q \wedge r$	A $p \Leftrightarrow q \wedge r$	$\sim r$	$\sim p$	B $\sim r \Rightarrow \sim p$	$A \Rightarrow B$
T	T	T	T	T	F	F	T	T
T	T	F	F	F	T	F	F	T
T	F	T	F	F	F	F	T	T
T	F	F	F	F	T	F	F	T
F	T	T	T	F	F	T	T	T
F	T	F	F	T	T	T	T	T
F	F	T	F	T	F	T	T	T
F	F	F	F	T	T	T	T	T

Since all entries of the last column are T . Hence the given formula is a tautology.

(iv) Proceed as above in (iii).

Example 15. Simplify the following:

Example 5. Show that $p \oplus q \equiv (p \wedge \sim q) \vee (\sim p \wedge q)$.

Solution. As in Ex. 3 above, we shall prove that

$p \oplus q \Rightarrow (p \wedge \sim q) \vee (\sim p \wedge q)$ is a tautology.

p	q	$p \oplus q$	$\sim p$	$\sim q$	$p \wedge \sim q$	$\sim p \wedge q$	$(p \wedge \sim q) \vee (\sim p \wedge q)$	$p \oplus q \Leftrightarrow (p \wedge \sim q) \vee (\sim p \wedge q)$
T	T	F	F	F	F	F	F	T
T	F	T	F	T	T	F	T	T
F	T	T	T	F	F	T	T	T
F	F	F	T	T	F	F	F	T

Example 6. Show that $(p \uparrow q) \oplus (p \uparrow q) \equiv (p \vee q) \wedge (p \downarrow q)$.

Solution. As in Ex. 3 above, we shall prove that

$(p \uparrow q) \oplus (p \uparrow q) \Rightarrow (p \vee q) \wedge (p \downarrow q)$ is a tautology.

p	q	$p \uparrow q$	$(p \uparrow q) \oplus (p \uparrow q)$	$p \wedge q$	$p \downarrow q$	$(p \vee q) \wedge (p \downarrow q)$	$(p \uparrow q) \oplus (p \uparrow q) \Leftrightarrow (p \vee q) \wedge (p \downarrow q)$
T	T	F	F	T	F	F	T
T	F	T	F	T	F	F	T
F	T	T	F	T	F	F	T
F	F	T	F	F	T	F	T

Example 7. Show that $(p \uparrow q) \oplus (p \uparrow q)$ is logically false (or is a contradiction).

Solution. In truth table of Ex. 6 above all entries in column four are of 'F' only. Hence $(p \uparrow q) \oplus (p \uparrow q)$ is logically false.

Example 8. Prove that following statements are logically equivalent :

- $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$.
- $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.
- $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.
- $p \Rightarrow (q \wedge r) \equiv (p \Rightarrow q) \wedge (p \Rightarrow r)$.
- $p \Rightarrow (q \vee r) \equiv (p \Rightarrow q) \vee (p \Rightarrow r)$.
- $(p \Rightarrow q) \vee r \equiv (p \vee r) \Rightarrow (q \vee r)$.
- $p \vee (q \Leftrightarrow r) \equiv (p \vee q) \Leftrightarrow (p \vee r)$.

[R.G.P.V. June 2007]

Solution. These are easily proved by truth tables.

Some Important Law,

1) Idempotent law $\rightarrow$

$$P \wedge P \iff P \quad \text{and} \quad P \vee P \iff P$$

2) Commutative law $\rightarrow$

$$P \wedge Q \iff Q \wedge P \quad \text{and} \quad P \vee Q \iff Q \vee P$$

3) Associative law $\rightarrow$

$$i) (P \vee Q) \vee R \iff P \vee (Q \vee R)$$

$$ii) (P \wedge Q) \wedge R \iff P \wedge (Q \wedge R)$$

4) Distributive law $\rightarrow$

$$i) P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R)$$

$$ii) P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$$

5) De-Morgan's law $\rightarrow$

$$i) \sim(P \vee Q) \iff (\sim P) \wedge (\sim Q)$$

$$ii) \sim(P \wedge Q) \iff (\sim P) \vee (\sim Q)$$

Normal form $\rightarrow$

Let $A (P_1, P_2, P_3, \dots, P_n)$ be a statement formula then the construction of truth table may not be practical always. So, we consider alternate procedure known as reduction to normal form.

i) Disjunction Normal Form $\rightarrow$ A formula (or statement) which ~~consist~~ consist of disjunction between conjunction is called Disjunction Normal form DNF.

Ex $\rightarrow$ 1 $(P \wedge Q) \vee R$

disjunction between two (equal)
 $(\wedge) \vee (\wedge)$

Ex $\rightarrow$ 2 $(P \wedge \sim Q) \vee (\sim P \wedge R) \vee (R \wedge \sim Q)$

Ex obtain the DNF of the form $(P \rightarrow Q) \wedge (\sim P \wedge Q)$

Sol $\rightarrow$ We know that

$$(P \rightarrow Q) \wedge (\sim P \wedge Q)$$

$$\text{So. } (\sim P \vee Q) \wedge (\sim P \wedge Q)$$

Apply distributive law.

$$(\sim P \wedge \sim P \wedge Q) \vee (Q \wedge \sim P \wedge Q)$$

$$(\sim P \wedge Q) \vee (Q \wedge \sim P)$$

$$\begin{aligned} & \left(\because \begin{array}{l} \text{(idempotent law)} \\ \sim P \wedge \sim P = \sim P \end{array} \right) \\ & \left(\because Q \wedge Q = Q \right) \end{aligned}$$

then table $\downarrow$

ii) Conjunction Normal Form $\rightarrow$

A statement form which consists of conjunction between disjunction is called CNF.

Ex $\rightarrow$ i) $P \wedge Q$

Ex $\rightarrow$ ii) $(\sim P \vee Q) \wedge (\sim P \vee R)$

①
conjunction between the
two disjunction
② $\wedge$ ③

Ex obtain CNF of the form $(P \wedge Q) \vee (\sim P \wedge Q \wedge R)$

Sol $\rightarrow$ $(P \wedge Q) \vee (\sim P \wedge Q \wedge R)$

using distributive law.

$$P \vee (\sim P \wedge Q \wedge R) \wedge Q \vee (\sim P \wedge Q \wedge R)$$

$$\{(P \vee \sim P) \wedge (P \vee Q) \wedge (P \vee R)\} \wedge \{(Q \vee \sim P) \wedge (Q \vee Q) \wedge P(Q \wedge R)\}$$

$$\begin{aligned} &(\because Q \vee Q = Q \text{ idempotent}) \\ &(\because P \vee \sim P = 1 \text{ negation}) \end{aligned}$$

$$(P \vee Q) \wedge (P \vee R) \wedge \{(Q \vee \sim P) \wedge Q \wedge (Q \wedge R)\} \quad \underline{\text{Ans}}$$

Ques ① obtain DNF of $P \vee (\sim P \rightarrow (Q \vee (Q \rightarrow \sim R)))$

Sol $\rightarrow$ $P \vee (\sim P \rightarrow (Q \vee (Q \rightarrow \sim R)))$

$$P \vee (\sim P \rightarrow (Q \vee (\sim Q \vee \sim R)))$$

$$P \vee (\sim P \rightarrow ((Q \vee \sim Q) \vee (Q \vee \sim R)))$$

$$P \vee (\sim P \rightarrow (Q \vee \sim R))$$

$$P \vee (P \vee (Q \vee \sim R))$$

$Q \rightarrow \sim R$
$\sim Q \vee \sim R$

$$(\because Q \vee \sim Q = 1)$$

$$p \vee ((p \vee q) \vee (p \vee \sim r))$$

$$p \vee (p \vee q \vee p \vee \sim r)$$

$$p \vee (p \vee q \vee \sim r)$$

$$(p \vee p) \vee (p \vee q) \vee (p \vee \sim r)$$

$$(p \vee q) \vee (p \vee \sim r)$$

$$(p \vee q \vee \sim r)$$

$$(\because p \vee p = p \text{ idempotent})$$

$$(\because \text{idempotent } p \vee p = p)$$

Ex 3 Obtain CNF of $(p \rightarrow q) \wedge (q \vee (p \wedge r))$
and determine whether or not it is tautology.

Sol We know that $(p \rightarrow q) \wedge (q \vee (p \wedge r))$
 $(\sim p \vee q) \wedge (q \vee p) \wedge (q \vee r)$

Truth table

p	q	r	$\sim p$	$\sim p \vee q$	$q \vee p$	$q \vee r$	$(\sim p \vee q) \wedge (q \vee p) \wedge (q \vee r)$
T	T	T	F	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	F	T	T	F
T	F	F	F	F	T	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	T	T	F	T	F
F	F	F	T	T	F	F	F

Ex → Find ^{CNF} ~~DNF~~ of $p \wedge (p \rightarrow q)$

Sol →

$$p \wedge (p \rightarrow q)$$

$$p \wedge (\sim p \vee q)$$

$$(p \wedge \sim p) \vee (p \wedge q)$$

$$p \wedge q$$

$$[\because (p \wedge \sim p) = 0]$$

Ex →

obtain CNF of $(\sim p \rightarrow r) \wedge (q \leftrightarrow p)$

Sol →

$$(\sim p \rightarrow r) \wedge (q \leftrightarrow p)$$

$$(\sim p \rightarrow r) \wedge (q \rightarrow p) \wedge (p \rightarrow q)$$

$$(p \vee r) \wedge (\sim q \vee p) \wedge (\sim p \vee q)$$

Find Truth table →

Ex → Find DNF of $p \vee \{ \sim p \rightarrow (q \vee (q \rightarrow \sim r)) \}$

$$\text{Ans} \rightarrow \{ p \vee q \vee \sim r \}$$

Ex →

Express the following formula into DNF and CNF

$$\sim(p \vee q) \leftrightarrow (p \wedge q)$$

Quantifiers

Quantifiers are the words that refer to Quantities such as "Some" or "all" it tells for how many elements a given Predicate is True.

Quantifiers are used to express the Quantities without giving an exact no.

The word or a symbol which convey the idea of Quantity or no. are called Quantifiers.

Ex → all, some, many, none, few etc.

Sentence like : "Can I have some water?"
"Jack has many friends here."

There are two most Important type of Quantifiers.

- 1) Universal Quantifiers
- 2) Existential Quantifiers.

1) Universal Quantifiers → It states that the Statement within its Scope are true "for every value" of the specific variable. It is denoted by symbol $\forall$

We write it as " $\forall x P(x)$ " which means that $P(x)$ is true for every x in the Domain.

$\forall x$ represent "for all x ", "for every x ", "for each x "

Ex $\rightarrow$ ① All Human Beings are mortal. — ①

$P(x) : x$ is mortal

eg ① can be written as

$$(\forall x \in U) P(x) \quad \text{or} \quad \forall x \cdot P(x)$$

where U - domain or universe of Discourse.

Ex $\rightarrow$ ② All students are Successful.

$P(x) : x$ is Successful

$$\forall x P(x).$$

Ex $\rightarrow$ Let $P(x)$ be a statement $x+1 > x$, where x is $\in \mathbb{Z}$

$$P(1) : 1+1 > 1 = 2 > 1 \quad (\text{True})$$

$$P(2) : 2+1 > 2 = 3 > 2 \quad (\text{True})$$

$$P(3) : 3+1 > 3 = 4 > 3 \quad (\text{True})$$

!

$P(x) : x+1 > x$ is true for all the integers

or

$\forall x P(x)$ is true.

Existential Quantifiers $\rightarrow$

If state that the statement within its scope are true for some values of the specific variable. it is denoted by $\exists$

We can write it as " $\exists x P(x)$ " which means that, $P(x)$ is true for some x in the Domain.

$\exists x$ represent "there exists an x "

"there is an x "

"for some x "

"there is at least one x "

Ex $\rightarrow$ ① Some Human Beings are Mortal

$P(x)$: x is mortal.

$\Rightarrow \exists x P(x)$.

Ex $\rightarrow$ ② Let $Q(x)$ be the statement $x < 2$, where x is the integers.

$Q(x)$: $x < 2$

$Q(1)$: $1 < 2$ (True)

$Q(2)$: $2 < 2$ (False)

$Q(3)$: $3 < 2$ (False)

Ex-3) Some students are successful

$P(x)$: x is Successful

$\exists x P(x)$

Ex-4) There exist a student

$P(x)$: x is student

$\exists x P(x)$

Negation of a Quantifiers $\rightarrow$

To Construct the Negation of a Quantified Statement. Change the Quantifier from Universal to Existential and Existential to Universal and also replace the Statement.

Universal Quantifier negation will be Existential Quantifier
Existential Quantifier negation will be Universal Quantifier

Statement

Negation

1) $\forall x P(x)$ (all true)

1) $\exists x [\neg P(x)]$ (at least one false)

2) $\exists x [\neg P(x)]$ ———

2) $\forall x P(x)$ ———

3) $\forall x [\neg P(x)]$ All false

3) $\exists x P(x)$ at least one true

4) $\exists x P(x)$ ———

4) $\forall x [\neg P(x)]$ ———

Negate the Statements

Ex-① All integers are greater than 8

$P(x)$: x is an integer

$Q(x)$: x is greater than 8

$$\forall x P(x) \rightarrow Q(x)$$

$$\underline{x} \equiv \forall x (\sim P(x) \vee Q(x))$$

Negation $\rightarrow$

$$\exists x \sim (\sim P(x) \vee Q(x))$$

$$\equiv \exists x (P(x) \wedge \sim Q(x))$$

Ex-② Not all men are genius

$M(x)$: x is a man, $G(x)$: x is genius

$$\sim [\forall x M(x) \rightarrow G(x)]$$

$$\equiv \forall x \sim (M(x) \rightarrow G(x))$$

Negation $\rightarrow$ $\equiv \exists x (M(x) \rightarrow G(x))$

$$(\forall x P(x) \vee (\exists y P(y)))$$

Ex-③

Negation $\rightarrow$ $\sim [\forall x P(x) \vee (\exists y P(y))]$

$$\equiv \exists x (\sim P(x)) \wedge \forall y (\sim P(y))$$

Ex-4 Some students are intelligent
 $S(x)$: x is a student , $I(x)$: x is intelligent

$$\exists x (S(x) \wedge I(x))$$

Negation $\rightarrow$ $\forall x \sim (S(x) \wedge I(x))$
 $\forall x \sim S(x) \vee \sim I(x)$

Ex-5 Any integer is either positive or negative
 $I(x)$: x is an integer , $P(x)$: x is either positive or negative

$$\forall x [I(x) \rightarrow P(x)]$$

$$\forall x [\sim I(x) \vee P(x)]$$

Negation $\rightarrow$ $\exists x [\sim I(x) \wedge \sim P(x)]$

$$\exists x [I(x) \wedge \sim P(x)]$$